MTH603

Assignment # 01 Solution 2023

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Question #01:

Find the roots of the given equation x^3-x-11=0, in the interval (2,3) using Bisection Method after five iterations.

Note: Accuracy up to four decimal places is required.

Solution:

Here $x^3 - x - 11 = 0$

let $f(x) = x^3 - x - 11$

- 1*st Iteration*:

Here $f(2) = -5 < 0$ *and* $f(3) = 13 > 0$

Now Root lies between 2 *and* 3

$$x_o = \frac{2+3}{2} = 2.5$$

$$f(x_o) = f(2.5) = (2.5)^3 - 2.5 - 11 = 2.125 > 0$$

- 2*nd Iteration*

Here $f(2) = -5 < 0$ *and* $f(2.5) = 2.125 > 0$

Now root lie between 2 *and* 2.5

$$x_1 = \frac{2+2.5}{2} = 2.25$$

$$f(x_1) = f(2.25) = 2.25^3 - 2.25 - 11 = -1.8594 < 0$$

•3rd Iteration:

Here $f(2.25) = -1.8594 < 0$ and $f(2.5) = 2.125 > 0$

Now root lies between 2.25 and 2.5

$$x_2 = \frac{2.25 + 2.5}{2} = 2.375$$

$$f(x_2) = f(2.375) = (2.375)^3 - 2.375 - 11 = 0.0215 > 0$$

•4th Iteration:

Here $f(2.25) = -1.8594 < 0$ and $f(2.375) = 0.0215 > 0$

Now, root lie between 2.25 and 2.375

$$x_3 = \frac{2.25 + 2.375}{2} = 2.3125$$

$$f(x_3) = f(2.3125) = (2.3125)^3 - 2.3125 - 11 = -0.946 < 0$$

•5th Iteration:

Here $f(2.3125) = -0.946 < 0$ and $f(2.375) = 0.0215 > 0$

Now, root lies between 2.3125 and 2.375

$$x_4 = \frac{2.3125 + 2.375}{2} = 2.3438$$

$$f(x_4) = f(2.3438) = (2.3438)^3 - 2.3438 - 11 = -0.4691 < 0$$

Approximate Root of given function is -0.4691 (After 5 Iteration).

Question#02:

Find a root of the given equation in the interval (1,2) using Newton-Raphson Method after four iterations:

$x^3 - 20 = -2x^2 - 10x$

take Initial value 1.2.

Solution:

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Here $x^3 - 20 = -2x^2 - 10x$

$x^3 + 2x^2 + 10x - 20 = 0$

Let $f(x) = x^3 + 2x^2 + 10x - 20$

$f'(x) = 3x^2 + 4x + 10$

$x_o = 1.2$

- 1*st Iteration*:

$f(x_o) = f(1.2) = 1.2^3 + 2(1.2)^2 + 10(1.2) - 20 = -3.392$

$f'(x_o) = f'(1.2) = 3(1.2)^2 + 4(1.2) + 10 = 19.12$

$$x_1 = x_o - \frac{f(x_o)}{f'(x_o)}$$

$$x_1 = 1.2 - \frac{-3.392}{19.12}$$

$x_1 = 1.3774$

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• *2nd Iteration*:

$$f(x_1) = f(1.3774) = (1.3774)^3 + 2(1.3774)^2 + 10(1.3774) - 20 = 0.1818$$

$$f'(x_1) = f'(1.3774) = 3(1.3774)^2 + 4(1.3774) + 10 = 21.2014$$

$$x_2 = x_1 - \frac{f(x_1)}{f(x_1)}$$

$$x_2 = 1.3774 - \frac{0.1818}{21.2014}$$

$$x_2 = 1.3688$$

• *3rd Iteration*:

$$f(x_2) = f(1.3688) = (1.3688)^3 + 2(1.3688)^2 + 10(1.3688) - 20 = 0.0005$$

$$f'(x_2) = f'(1.3688) = 3(1.3688)^2 + 4(1.3688) + 10 = 21.0964$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$x_3 = 1.3688 - \frac{0.0005}{21.0964}$$

$$x_3 = 1.3688$$

• *4th Iteration*:

$$f(x_3) = f(1.3688) = (1.3688)^3 + 2(1.3688)^2 + 10(1.3688) - 20 = 0.0005$$

$$f'(x_3) = f'(1.3688) = 3(1.3688)^2 + 4(1.3688) + 10 = 21.0964$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$$

$$x_4 = 1.3688 - \frac{0.0005}{21.0964}$$

$$x_4 = 1.3688$$

Approximate root of the equation $x^3 + 2x^2 + 10x - 20 = 0$ *u sin g newton raphson method is* 1.3688 *(after 4 iterations)*

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